■■ C printf() Cheat Sheet

■ Common Format Specifiers

Type / Meaning	Format	Example	Output
Integer (signed)	%d / %i	printf("%d", -42);	-42
Unsigned int	%u	printf("%u", 42);	42
Hexadecimal	%x / %X	printf("%x", 255);	ff
Octal	%o	printf("%o", 8);	10
Long int	%ld	printf("%ld", 123456L);	123456
Float / double	%f	printf("%f", 3.14);	3.140000
Scientific notation	%e / %E	printf("%e", 3.14);	3.140000e+00
Character	%c	printf('%c', 'A');	A
Pointer / address	%p	printf('%p', ptr);	0x7ffeea2c
Percent sign	%%	printf("%%");	%

■ Width, Precision & Alignment

Format	Meaning	Example	Output
%5d	Width 5 (right align)	printf("[%5d]", 42);	[42]
%-5d	Left align	printf("[%-5d]", 42);	[42]
%05d	Zero-pad	printf("[%05d]", 42);	[00042]
%.2f	2 decimals	printf("%.2f", 3.14159);	3.14
%8.3f	Width 8, 3 decimals	printf("[%8.3f]", 3.14);	[3.140]

■ Mixing Text & Variables

```
int a = 10;
float b = 2.718;

printf("a = %d, b = %.2f\n", a, b);
// Output: a = 10, b = 2.72
```